

# Microeconomics

## Lecture 12: Firm Supply and Market Supply

Marco Rosso

University of Bologna

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## Recap: What We Already Know

From the previous lecture we know that the price-taking firm chooses the quantity that maximizes:

$$\pi(Q) = PQ - C(Q)$$

### Key results already established:

- First-order condition:

$$P = MC(Q^*)$$

- Second-order condition:

$$MC'(Q^*) > 0$$

- In the short run the firm produces only if:

$$P \geq AVC_{\min}$$

## Recap: Yesterday's Key Point

Yesterday we answered this question:

### Question from L11

**Given a market price  $P$ , how much should the firm produce to maximize profit?**

The answer is:

- find the point where  $P = MC$
- choose the portion where  $MC$  is increasing
- check the shutdown condition

So yesterday we found a **single optimal quantity**:

$Q^*$

for a given price.

# Today's New Question

Today we take a step forward.

## New question

If the price **changes**, how does the firm's optimal quantity change?

In other words:

- yesterday: given  $P$ , we found  $Q^*$
- today: we let  $P$  vary and build a **supply function**

This function will then be the basis for moving on to **market supply**.

# Learning Objectives

After this lecture you will be able to:

1. **Define** the individual supply curve as the set of optimal choices as  $P$  varies
2. **Derive**  $Q^s(P)$  analytically by inverting the condition  $P = MC$
3. **Distinguish** a movement along the curve from a shift of the curve
4. **Analyze** how changes in costs shift the supply curve
5. **Build** the market supply curve by horizontal aggregation
6. **Aggregate** the supply of heterogeneous firms while respecting the *shutdown condition*
7. **Compute** the elasticity of supply and interpret its determinants
8. **Solve** complete numerical exercises on individual supply and aggregation

# From Optimal Choice to Supply Function

In the previous lecture we solved:

$$\max_{Q \geq 0} \pi(Q) = PQ - C(Q) \quad \Rightarrow \quad P = MC(Q^*)$$

But there  $P$  was treated as **given**.

## Today's conceptual step

Now imagine that  $P$  can take different values:

$$P_1, P_2, P_3, \dots$$

For each price, the firm chooses a different optimal quantity:

$$Q^*(P_1), Q^*(P_2), Q^*(P_3), \dots$$

# The Supply Curve as a Collection of Optima

Putting together all the pairs

$$(P, Q^*(P))$$

we obtain a new relationship between price and quantity.

## Central idea

The **individual supply curve** is the collection of all the optimal quantities the firm chooses for all possible prices.

So the supply curve:

- derives from the rule  $P = MC$
- is not a technological relationship
- is not a curve chosen a priori
- is the result of a profit-maximization problem

# Operational Definition of the Individual Supply Curve

From Lecture 11 we know that the firm produces in the short run only if:

$$P \geq AVC_{\min}$$

and, when it produces, it chooses the quantity such that:

$$P = MC(Q^*) \quad \text{with} \quad MC'(Q^*) > 0$$

## Operational definition

The short-run individual supply curve is:

$$Q^s(P) = \begin{cases} Q : MC(Q) = P, MC'(Q) > 0 & \text{if } P \geq AVC_{\min} \\ 0 & \text{if } P < AVC_{\min} \end{cases}$$

## Economic Translation of the Definition

The previous definition contains three simple economic ideas.

1. **Optimality:** the quantity must satisfy  $P = MC$
2. **Maximum, not minimum:** we choose the portion where  $MC$  is increasing
3. **Shutdown:** below  $AVC_{\min}$  the firm supplies nothing

### Conclusion

The individual supply curve **does not coincide with the whole  $MC$** , but only with the economically relevant portion.

# Not All of $MC$ Is Supply

This is one of the most frequent confusions.

## Typical mistake

“The supply curve coincides with the whole marginal cost curve.”

## Why is it false?

- in the portion below  $AVC_{\min}$  the firm prefers to shut down
- so those points of  $MC$  do not represent quantities actually supplied
- only above the shutdown threshold does  $MC$  become the supply curve

## Message to remember:

SR supply = increasing portion of  $MC$  above  $AVC_{\min}$

# Direct Form and Inverse Form

The same supply curve can be written in two ways.

## Direct form

$$Q^s = Q^s(P)$$

Given the price, we read off the quantity supplied.

## Inverse form

$$P = P^s(Q)$$

Given the quantity, we read off the minimum price required.

## Practical use:

- the direct form is useful in numerical exercises
- the inverse form is useful in graphs with  $P$  on the vertical axis
- in inverse form, supply coincides with the relevant  $MC$

## Simple Example of Direct and Inverse Form

If

$$MC = 2Q + 4$$

the optimality condition is:

$$P = 2Q + 4$$

**Direct form:**

$$Q^s(P) = \frac{P - 4}{2} \quad \text{for } P \geq 4$$

**Inverse form:**

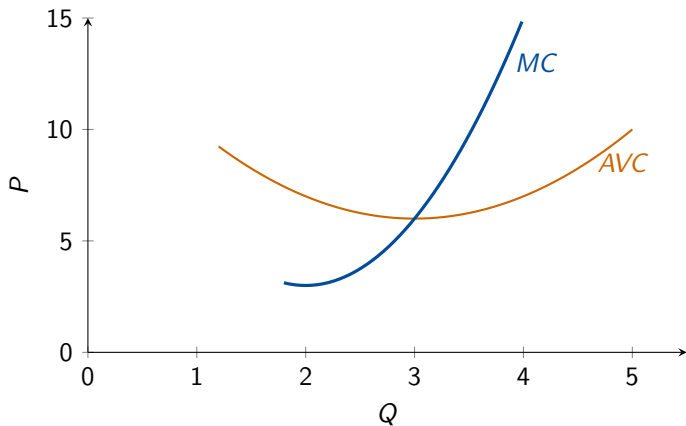
$$P = 2Q + 4$$

The two expressions describe the **same curve**, simply read in two different directions.

## Graphical Construction: Starting Point

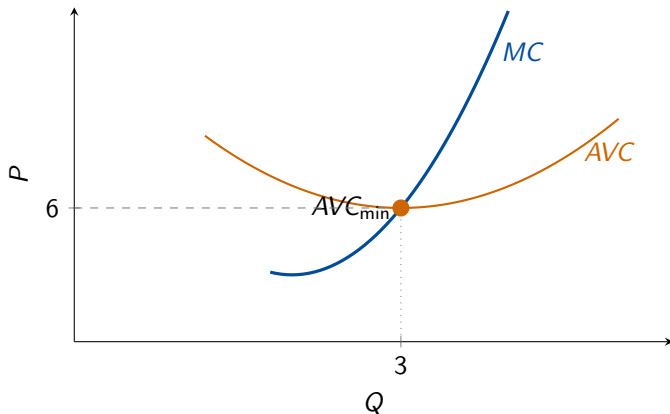
Before building the supply curve, let us look at the two relevant curves:

- $AVC$
- $MC$



## Graphical Construction: Identifying the Shutdown Threshold

The first thing to do is to find the minimum of  $AVC$ .

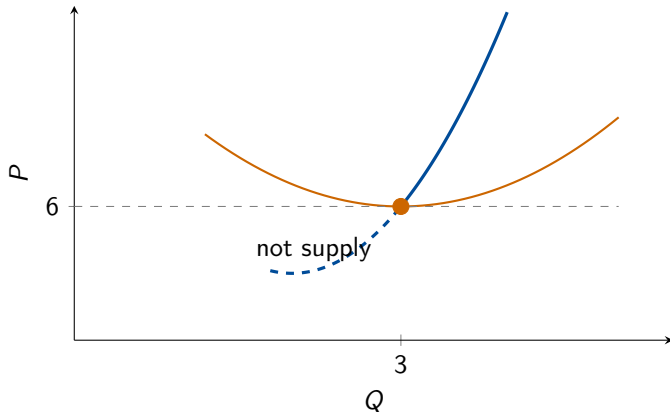


Here the shutdown threshold is:

$$AVC_{\min} = 6$$

## Graphical Construction: The Portion Below the Threshold

What happens at prices below  $AVC_{\min}$ ?

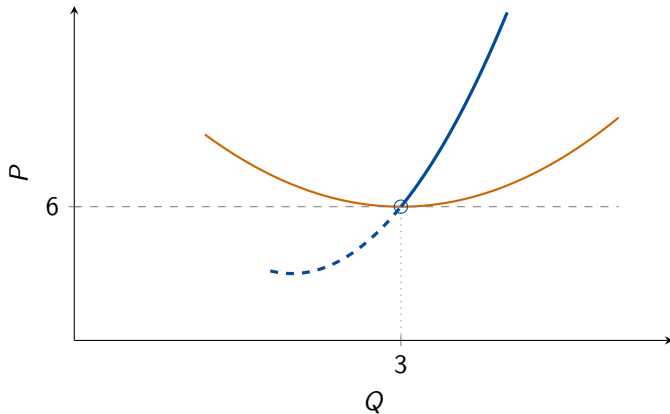


The portion of  $MC$  below  $AVC_{\min}$  **is not supply**: at those prices the firm does not produce.

## Graphical Construction: Fix a First Price

Now suppose the price is:

$$P = 6$$



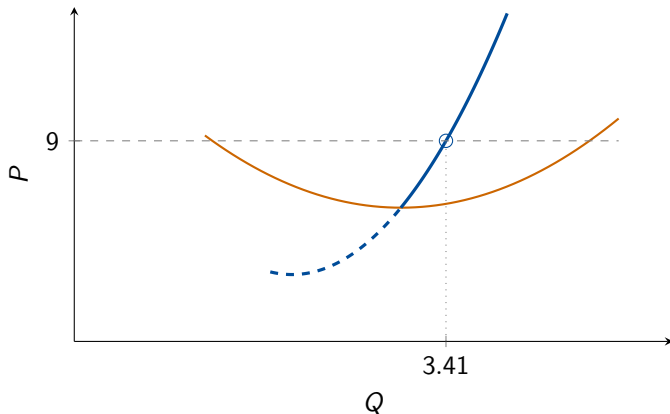
The optimal point is:

$$Q^* = 3$$

## Graphical Construction: A Higher Price

Now suppose the price rises to:

$$P = 9$$



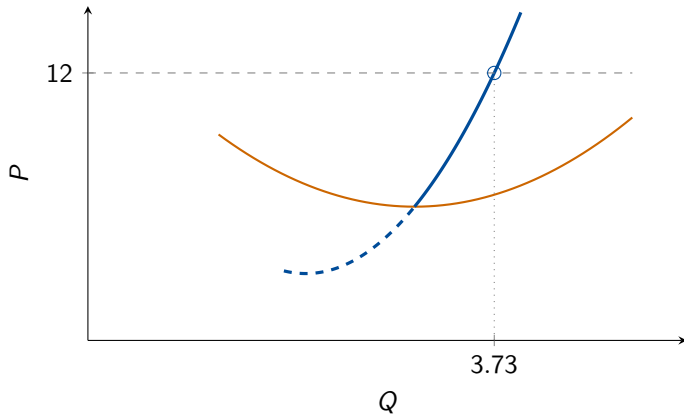
The new optimal point is:

$$Q^* \approx 3.41$$

## Graphical Construction: An Even Higher Price

If the price rises further to:

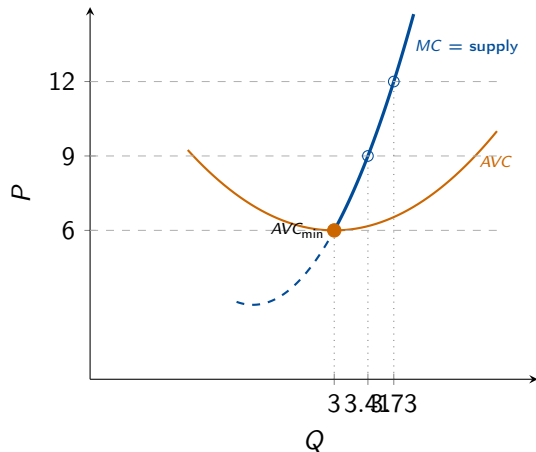
$$P = 12$$



Now the optimal quantity becomes:

$$Q^* \approx 3.73$$

# Graphical Construction: Putting the Points Together



Each price generates a different optimal point.

## Reading off the three prices:

- $P = 6$ :  $Q^* = 3$
- $P = 9$ :  $Q^* \approx 3.41$
- $P = 12$ :  $Q^* \approx 3.73$

By letting  $P$  vary, we obtain the supply curve.

The solid thick portion of  $MC$  is the individual supply curve.

# Why Is the Supply Curve Increasing?

The answer lies in the second-order condition.

If the optimum requires:

$$MC'(Q^*) > 0$$

then marginal cost is increasing in the relevant portion.

## Intuition

If the price rises, a larger quantity is needed to match a higher level of  $P$ , because  $MC$  is increasing.

Therefore:

$$P \uparrow \Rightarrow Q^s \uparrow$$

## Summary message

The supply curve is increasing because it inherits the increasing portion of the marginal cost curve.

## Analytical Derivation: Setup

Consider the example again:

$$C(Q) = Q^3 - 6Q^2 + 15Q + 10$$

Then:

$$MC = 3Q^2 - 12Q + 15$$

and from Lecture 11 we know that:

$$AVC_{\min} = 6$$

Now we want to obtain the **supply curve** in analytical form:

$$Q^s(P)$$

## Analytical Derivation: Set $P = MC$

We start from the optimality condition:

$$P = MC$$

that is:

$$P = 3Q^2 - 12Q + 15$$

We rewrite everything in quadratic form:

$$Q^2 - 4Q + \frac{15 - P}{3} = 0$$

Now we solve this equation for  $Q$ .

## Analytical Derivation: Solve the Equation

Applying the quadratic formula we obtain:

$$Q = 2 \pm \sqrt{4 - \frac{15 - P}{3}}$$

that is:

$$Q = 2 \pm \sqrt{\frac{P - 3}{3}}$$

So there are **two candidate solutions**:

$$Q_1 = 2 - \sqrt{\frac{P - 3}{3}} \quad Q_2 = 2 + \sqrt{\frac{P - 3}{3}}$$

## Analytical Derivation: Select the Correct Branch

Not all solutions of  $P = MC$  are profit-maximizing points.

We must choose the branch with:

$$MC'(Q) > 0$$

In our example:

$$MC'(Q) = 6Q - 12$$

This is positive only for:

$$Q > 2$$

So we choose the branch:

$$Q = 2 + \sqrt{\frac{P - 3}{3}}$$

## Analytical Derivation: Final Supply Curve

At this point we also impose the shutdown condition:

$$P \geq 6$$

The individual supply curve is therefore:

$$Q^s(P) = 2 + \sqrt{\frac{P-6}{3}}, \quad P \geq 6$$

and:

$$Q^s(P) = 0 \quad \text{if } P < 6$$

Check

$$P = 6 \Rightarrow Q^s = 3$$

$$P = 15 \Rightarrow Q^s = 4$$

## Linear Case: A Simpler Supply

Now consider:

$$C(Q) = Q^2 + 4Q + 9$$

Then:

$$MC = 2Q + 4$$

Setting:

$$P = 2Q + 4$$

we immediately obtain:

$$Q^s(P) = \frac{P - 4}{2} \quad \text{for } P \geq 4$$

This case is particularly useful because:

- the inversion is immediate
- the supply curve is linear
- calculations in exercises are simpler

# Operational Checklist: How to Build a Supply Curve

Given a cost function  $C(Q)$ :

1. compute marginal cost

$$MC(Q) = \frac{dC(Q)}{dQ}$$

2. compute average variable cost

$$AVC(Q) = \frac{VC(Q)}{Q}$$

3. find the shutdown threshold

$$AVC_{\min}$$

4. solve the condition

$$P = MC(Q)$$

5. select the branch with

$$MC'(Q) > 0$$

6. finally impose

$$P \geq AVC_{\min}$$

## Typical Mistake to Avoid

Once this step is done:

$$P = MC(Q) \Rightarrow Q^*$$

You must not stop there.

### Mistake

Finding  $Q^*$  for a given price is not yet the same as writing the supply curve.

To obtain the supply curve you must:

- treat  $P$  as a variable
- derive  $Q$  as a function of  $P$
- add the shutdown condition

## Supply Curve Table: Linear Example

With

$$Q^s(P) = \frac{P - 4}{2}$$

the firm produces for any:

$$P \geq 4$$

| $P$ | $Q^*$ | $AVC(Q^*)$ | $AC(Q^*)$ | $\pi$ |
|-----|-------|------------|-----------|-------|
| 4   | 0     | 4.0        | —         | −9    |
| 6   | 1     | 5.0        | 14.0      | −8    |
| 8   | 2     | 6.0        | 10.5      | −5    |
| 10  | 3     | 7.0        | 10.0      | 0     |
| 12  | 4     | 8.0        | 10.25     | +7    |
| 14  | 5     | 9.0        | 10.8      | +16   |

## Reading the Table

The previous table shows three economically different zones.

- $P = 4$ : shutdown point

$$Q^* = 0, \quad \pi = -FC = -9$$

- $P = 10$ : long-run break-even point

$$\pi = 0, \quad P = AC_{\min}$$

- $P = 12, 14$ : positive profit

$$\pi > 0$$

So the supply curve tells us how much quantity is produced, but the firm's situation also depends on:

- $AVC$
- $AC$
- profit

# Movement Along the Curve vs. Shift

## Movement along the curve

**Cause:** the price of the good  $P$  changes.

**Effect:** the optimal quantity changes on the **same** supply curve.

## Shift of the curve

**Cause:** costs or technology change.

**Effect:** the quantity supplied changes at **every** price.

## Movement Along the Curve: Example

Suppose the price rises from:

$$P = 8 \quad \text{to} \quad P = 12$$

and that the supply curve is:

$$Q^s = \frac{P - 4}{2}$$

Then:

$$Q^s(8) = 2 \qquad Q^s(12) = 4$$

### Interpretation

The supply curve has not changed. Only the point where we are **along the same curve** has changed.

## Shift of the Curve: Example

Now suppose the price of the good stays the same, but wages increase.

If marginal cost changes from:

$$MC = 2Q + 4 \quad \text{to} \quad MC' = 2Q + 7$$

then the supply curve changes too.

Before:

$$Q^s = \frac{P - 4}{2}$$

After:

$$Q^{s'} = \frac{P - 7}{2}$$

At any given price, the firm supplies less.

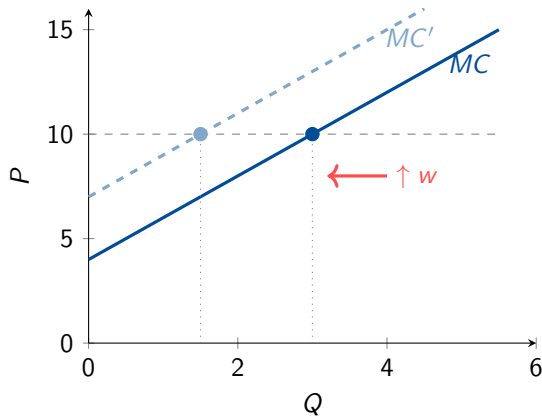
## Causes of Movements and Causes of Shifts

| Change                  | Movement along $S$ | Shift of $S$   |
|-------------------------|--------------------|----------------|
| Price of the good $P$   | ✓                  |                |
| Input prices ( $w, r$ ) |                    | ✓              |
| Technology              |                    | ✓              |
| Specific tax $t$        |                    | ✓              |
| Fixed cost              |                    | No (in the SR) |

### Key message

The price of the good changes the **chosen point** on the curve. Costs change the **curve itself**.

## Effect of an Increase in Variable Costs



**Scenario:** wages increase.

At a fixed price  $P = 10$ :

• before:

$$Q^* = 3$$

• after:

$$Q^{*'} = 1.5$$

The supply curve shifts to the left.

# Why Does an Increase in Variable Costs Shift Supply?

The logic is simple.

If variable costs increase:

- each additional unit costs more to produce
- marginal cost rises
- a higher price is needed to justify the same quantity

## Consequence

At any given price, the firm produces less:

$S$  shifts to the left

## Effect of Fixed Costs: Essential Reminder

**Frequent question:** “If rent increases, does the firm supply less?”

**In the short run: No.**

- fixed costs do not enter the marginal decision
- $MC$  does not change
- so the individual SR supply curve does not shift

**In the long run:** market supply can change, because some firms may exit.

# Individual Supply and Market Supply

So far we have studied the **individual firm**. Now we move to the market, which contains many firms.

## Basic idea

Market supply is obtained by summing, at each price, the quantities supplied by all active firms.

$$Q_{\text{mkt}}^S(P) = \sum_{i=1}^n Q_i^S(P)$$

# Why Do We Speak of Horizontal Aggregation?

In perfect competition the price is the same for all firms.

Therefore:

- we fix a common price  $P$
- each firm observes that price
- each firm chooses its own optimal quantity
- the market sums all these quantities

## Caution

We are not summing prices. We are summing the **quantities supplied at a common price**.

# Horizontal Aggregation: Procedure

To build the market supply curve:

1. fix a price  $P$
2. read off  $Q_i^s(P)$  for each firm
3. sum all the quantities

$$Q_{\text{mkt}}^s(P) = \sum_i Q_i^s(P)$$

4. repeat for many prices

## Key message

The market curve is the horizontal sum of the individual supply curves.

## Simple Case: $n$ Identical Firms

If all firms have the same supply curve:

$$Q^s(P)$$

and there are  $n$  of them, then:

$$Q_{\text{mkt}}^s(P) = n \cdot Q^s(P)$$

This is the simplest case of aggregation.

### Important consequence

The market curve has the same price threshold as individual supply, but a larger total quantity at any given price.

## Example with Identical Firms: Formula

Suppose each firm has:

$$Q_i^s = \frac{P - 4}{2}$$

and that there are 10 identical firms in the market.

Then:

$$Q_{\text{mkt}}^s = 10 \cdot \frac{P - 4}{2} = 5(P - 4) = 5P - 20 \quad \text{for } P \geq 4$$

Inverse form:

$$P = \frac{Q_{\text{mkt}}}{5} + 4$$

## Example with Identical Firms: Reading at a Given Price

Take:

$$P = 10$$

For the individual firm:

$$Q_i^s = \frac{10 - 4}{2} = 3$$

With 10 firms:

$$Q_{\text{mkt}}^s = 10 \cdot 3 = 30$$

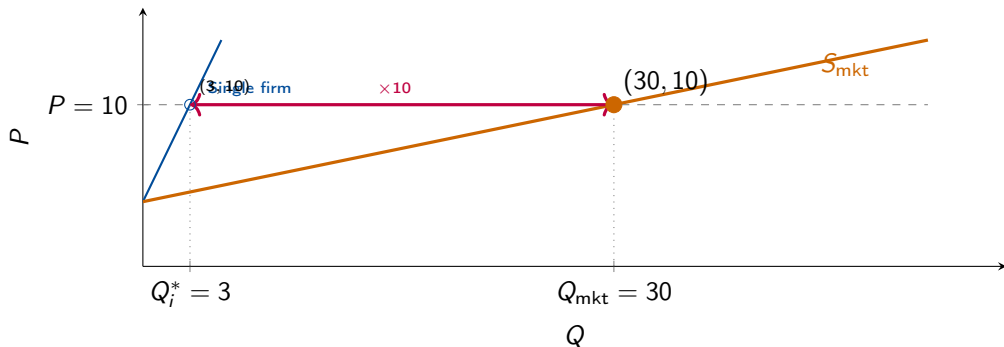
### Interpretation

Each firm reacts to the same price, but the total market quantity is the sum of the individual reactions.

# Graphical Aggregation: $n$ Identical Firms

Example: 10 identical firms with

$$Q_i^s = \frac{P - 4}{2}$$



## Why Is the Market Curve Flatter?

Suppose the price rises slightly.

**For the individual firm:** the quantity supplied rises a little, because the point where  $P = MC$  changes.

**For the market:** many changes happen simultaneously:

- each active firm supplies a little more
- any marginal firms may start producing
- if there are different groups of firms, new producers may enter

## Consequence: Greater Elasticity of Market Supply

Since, when the price rises, new firms may also start producing in addition to the active ones, the total market quantity tends to react, in percentage terms, more than the quantity of the individual firm.

### Consequence

The market supply curve is generally **more elastic** than the individual one.

Note: the market curve is in any case flatter than the individual one due to horizontal summation alone, but with  $n$  identical firms and no entry  $Q_{\text{mkt}}^S = n \cdot Q^S$  has the **same** elasticity as individual supply.

# Heterogeneous Firms: The General Case

Firms are not always identical.

They may differ in:

- marginal costs
- efficiency
- technology
- shutdown threshold

In this case aggregation requires more care, because not all firms produce at the same prices.

## Aggregation with Heterogeneous Firms: Example

Suppose the market contains:

- 50 type-A firms with

$$MC_A = 2Q_A \quad \Rightarrow \quad Q_A^s = \frac{P}{2}$$

- 30 type-B firms with

$$MC_B = 4Q_B \quad \Rightarrow \quad Q_B^s = \frac{P}{4}$$

Both types have:

$$AVC_{\min} = 0$$

## Aggregation with Heterogeneous Firms: Market Formula

Since both types produce for every  $P > 0$ , market supply is:

$$Q_{\text{mkt}}^S(P) = 50 \cdot \frac{P}{2} + 30 \cdot \frac{P}{4}$$

that is:

$$Q_{\text{mkt}}^S(P) = 25P + 7.5P = 32.5P$$

This is the market supply curve in the heterogeneous case.

## Aggregation with Heterogeneous Firms: Table

| $P$ | $Q_A^s$ (each) | $Q_B^s$ (each) | $Q_{A,tot}$ | $Q_{mkt}^s$ |
|-----|----------------|----------------|-------------|-------------|
| 4   | 2              | 1              | 100         | 130         |
| 8   | 4              | 2              | 200         | 260         |
| 12  | 6              | 3              | 300         | 390         |

The logic is always the same:

- first we find the supply of each type
- then we multiply by the number of firms
- finally we sum

# When Do Kinks Appear?

Kinks appear when different groups of firms start producing at different prices.

## Economic idea

At low prices only the most efficient firms produce.

When the price exceeds a certain threshold, firms with higher costs also start producing.

From that point on, market quantity grows more rapidly.

# Economic Interpretation of the Kink

The kink signals a change in the composition of the market.

## **Before the kink:**

- only one group of firms reacts

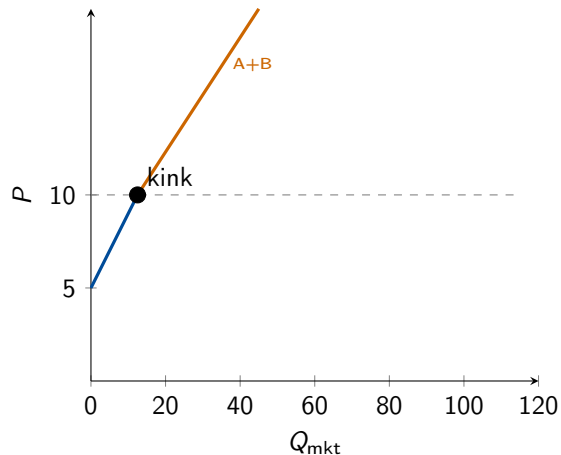
## **After the kink:**

- several groups react simultaneously
- total supply grows more rapidly

## Key message

The kink is the graphical reflection of the entry of new types of producers.

## Kink in the Market Curve: Graph



**Before the kink** ( $P < 10$ ): only A firms produce.

**At the kink** ( $P = 10$ ): B firms also enter.

**After the kink**: both groups react, so the curve becomes flatter.

# Elasticity of Supply: Definition

## Price elasticity of supply

$$\varepsilon_S = \frac{dQ^S}{dP} \cdot \frac{P}{Q^S} = \frac{\% \Delta Q^S}{\% \Delta P} \geq 0$$

### Interpretation:

$$\varepsilon_S = 2$$

means that a 1% increase in the price causes a 2% increase in the quantity supplied.

# How to Read Elasticity Economically

The elasticity of supply measures the **responsiveness** of quantity supplied to price.

- if  $\varepsilon_S$  is high, small price changes generate large quantity changes
- if  $\varepsilon_S$  is low, even large price changes generate small quantity changes

## Intuitive idea

The easier it is to expand production, the more elastic supply is.

## Classification of the Elasticity of Supply

| $\varepsilon_S$         | Term                | Curve      |
|-------------------------|---------------------|------------|
| 0                       | Perfectly inelastic | Vertical   |
| $0 < \varepsilon_S < 1$ | Inelastic           | Steep      |
| 1                       | Unit elastic        |            |
| $\varepsilon_S > 1$     | Elastic             | Flat       |
| $\rightarrow \infty$    | Perfectly elastic   | Horizontal |

## Elasticity of Supply: Numerical Example

Suppose that:

$$Q^s = 3P - 12 \quad \text{for } P \geq 4$$

Then:

$$\frac{dQ^s}{dP} = 3$$

At  $P = 8$ :

$$Q^s = 12 \quad \Rightarrow \quad \varepsilon_S = 3 \cdot \frac{8}{12} = 2$$

At  $P = 12$ :

$$Q^s = 24 \quad \Rightarrow \quad \varepsilon_S = 3 \cdot \frac{12}{24} = 1.5$$

# What Do We Observe in the Example?

In the previous example:

$$\varepsilon_S(8) = 2 \quad \text{and} \quad \varepsilon_S(12) = 1.5$$

So, even though the slope of the curve is constant, elasticity changes along the curve.

## Important message

Slope and elasticity are not the same thing.

Elasticity depends on:

- the derivative  $dQ/dP$
- the ratio of price to quantity

## Special Case: Straight Line Through the Origin

If the supply curve is:

$$Q^s = aP$$

then:

$$\varepsilon_S = a \cdot \frac{P}{aP} = 1$$

### Result

All linear supply curves through the origin have unit elasticity at every point.

# Limiting Cases

**Vertical curve:**

$$Q^s = \bar{Q} \quad \Rightarrow \quad \varepsilon_S = 0$$

**Horizontal curve:**

$$P = \bar{P} \quad \Rightarrow \quad \varepsilon_S \rightarrow \infty$$

These are the two extremes:

- perfectly inelastic supply
- perfectly elastic supply

# Determinants of the Elasticity of Supply

## When is supply highly elastic?

- when production can be expanded easily
- when inputs are easy to find
- when there is unused capacity
- when new firms can enter quickly

## When is supply not very elastic?

- when capacity is almost saturated
- when inputs are scarce or specialized
- when  $MC$  rises rapidly
- when it takes a long time to expand production

## Short Run vs Long Run

One of the most important determinants of elasticity is the time horizon.

- **Short run:** some inputs are fixed, so supply reacts less
- **Long run:** all inputs are variable and new firms can enter, so supply reacts more

### General rule

Supply tends to be more elastic in the long run than in the short run.

# Conceptual Checklist Before the Exercises

## To remember

To solve today's exercises correctly you need to keep four steps distinct.

1. **Individual firm:** given  $P$ , find  $Q^*$  with  $P = MC$
2. **Shutdown:** check whether the price is at least equal to  $AVC_{\min}$
3. **Supply curve:** turn the single optimum into a function  $Q^s(P)$
4. **Market:** sum the individual supplies horizontally

## Typical Mistakes to Avoid

1. “All of  $MC$  is supply”  
No: only the increasing portion above  $AVC_{\min}$ .
2. “If  $P < AC$ , the firm closes immediately”  
No: in the SR it still produces if  $P \geq AVC_{\min}$ .
3. “If  $P$  rises, the curve shifts”  
No: only the point along the curve changes.
4. “If fixed costs rise, individual SR supply shifts”  
No: the marginal decision does not change.

## Exercise 1

A firm has

$$C(Q) = Q^2 + 4Q + 9$$

1. Find  $MC$ ,  $AVC$ ,  $AVC_{\min}$ ,  $AC_{\min}$ .
2. Derive the supply curve  $Q^s(P)$ .
3. Build the supply table for

$$P \in \{4, 6, 8, 10, 12, 14\}$$

4. At  $P = 10$ : compute  $Q^*$  and  $\pi$ . Does the firm produce in the LR?

# Solution

1.

$$MC = 2Q + 4$$

$$VC = Q^2 + 4Q \quad \Rightarrow \quad AVC = Q + 4$$

AVC is increasing, so

$$AVC_{\min} = 4$$

Moreover

$$AC = Q + 4 + \frac{9}{Q}$$

$$\frac{dAC}{dQ} = 1 - \frac{9}{Q^2} = 0 \quad \Rightarrow \quad Q = 3$$

$$AC_{\min} = AC(3) = 10$$

## Solution

### 2. FOC:

$$P = MC = 2Q + 4 \quad \Rightarrow \quad Q = \frac{P - 4}{2}$$

Since  $MC' = 2 > 0$  always,

$$Q^s(P) = \frac{P - 4}{2}, \quad P \geq 4$$

### 3. Table:

| $P$ | $Q^*$ | $\pi = PQ^* - C(Q^*)$ | LR          |
|-----|-------|-----------------------|-------------|
| 4   | 0     | -9                    | Exits       |
| 6   | 1     | -8                    | Exits       |
| 8   | 2     | -5                    | Exits       |
| 10  | 3     | 0                     | Indifferent |
| 12  | 4     | +7                    | Stays       |
| 14  | 5     | +16                   | Stays       |

## Solution

4. At  $P = 10$ :

$$Q^* = 3$$

$$\pi = 10 \cdot 3 - (9 + 12 + 9) = 0$$

Since

$$P = AC_{\min} = 10$$

we are at the long-run break-even point.

### Interpretation

In the long run the firm is indifferent between staying and exiting.

## Exercise 2

The market contains 50 type-A firms with

$$C_A = Q_A^2 \quad \Rightarrow \quad MC_A = 2Q_A, \quad Q_A^s = \frac{P}{2}$$

and 30 type-B firms with

$$C_B = 2Q_B^2 \quad \Rightarrow \quad MC_B = 4Q_B, \quad Q_B^s = \frac{P}{4}$$

Both types have  $AVC_{\min} = 0$ .

1. Find the market supply curve  $Q^S(P)$ .
2. Compute  $Q^S$  for  $P = 4$ ,  $P = 8$ ,  $P = 12$ .
3. Write the inverse form.
4. If the number of A firms rises to 100, how does the market curve change?

## Solution

1.

$$Q^S = 50 \cdot \frac{P}{2} + 30 \cdot \frac{P}{4} = 25P + 7.5P = \boxed{32.5P}$$

2.

$$Q^S(4) = 130, \quad Q^S(8) = 260, \quad Q^S(12) = 390$$

3.

$$P = \frac{Q^S}{32.5}$$

## Solution

4. With 100 A firms:

$$Q^S = 100 \cdot \frac{P}{2} + 30 \cdot \frac{P}{4} = 50P + 7.5P = 57.5P$$

Therefore:

- the market curve shifts to the right
- at every price total supply increases
- the curve becomes flatter

## Exercise 3

A firm has the supply curve

$$Q^s = 4P - 8 \quad \text{for } P \geq 2$$

1. Compute  $\varepsilon_S$  at  $P = 4$  and  $P = 8$ .
2. How does elasticity change as  $P$  increases?
3. If the curve were

$$Q^s = 4P$$

what would  $\varepsilon_S$  be?

## Solution

1.

$$\frac{dQ^s}{dP} = 4$$

At  $P = 4$ :

$$Q^s = 8 \quad \Rightarrow \quad \varepsilon_S = 4 \cdot \frac{4}{8} = 2$$

At  $P = 8$ :

$$Q^s = 24 \quad \Rightarrow \quad \varepsilon_S = 4 \cdot \frac{8}{24} = \frac{4}{3} \approx 1.33$$

## Solution

2.

$$\varepsilon_S = 4 \cdot \frac{P}{4P - 8} = \frac{P}{P - 2}$$

As  $P$  increases, elasticity decreases and tends to 1.

3. If

$$Q^S = 4P$$

then

$$\varepsilon_S = 4 \cdot \frac{P}{4P} = 1$$

for every  $P$ .

# General Summary

- Individual supply curve: increasing portion of  $MC$  above  $AVC_{\min}$
- It is derived from  $P = MC(Q^*)$  by selecting the branch with  $MC'(Q^*) > 0$
- If  $P$  changes: movement along the supply curve
- If variable costs change: shift of the supply curve
- Fixed costs do not shift short-run individual supply
- Market supply: horizontal sum of individual supplies
- With heterogeneous firms the market curve may have kinks
- Elasticity of supply: measures the responsiveness of  $Q^S$  to changes in  $P$

## Skills Acquired

- ✓ Define the individual supply curve starting from  $MC$  and  $AVC$
- ✓ Derive  $Q^s(P)$  analytically from the condition  $P = MC(Q)$
- ✓ Distinguish movements along the supply curve from shifts of the curve
- ✓ Assess the effect of changes in variable costs and fixed costs
- ✓ Build the market supply curve by horizontal aggregation
- ✓ Handle cases with heterogeneous firms and kinked market curves
- ✓ Compute and interpret the elasticity of supply

## Optional Homework Exercises

1.

$$C(Q) = 3Q^2 + 6Q + 12$$

Find  $Q^s(P)$ ,  $AVC_{\min}$ ,  $AC_{\min}$ . Compute  $Q^s$  and  $\pi$  for  $P = 12, 18, 24, 30$ .

2. 60 type-A firms ( $MC_A = 5Q_A$ ) and 40 type-B firms ( $MC_B = 10Q_B$ ). Find  $Q^s(P)$ . Compute  $Q^s$  for  $P = 5, 10, 20$ .

3.

$$Q^s = 2P - 6 \quad \text{for } P \geq 3$$

Compute  $\varepsilon_S$  at  $P = 5$  and  $P = 10$ . How does elasticity change as  $P$  increases?

4. Explain why an increase in the profit tax does not shift the SR supply curve, whereas an increase in the cost of raw materials does.

5. A firm has

$$Q^s = \frac{P}{3}$$

If there are 90 identical firms in the market: write  $Q^s(P)$ . Find the price at which  $Q^s = 300$ .